

Team Presentation

Team 5

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Problem 1

Design two nonlinear algebraic equations (degree at least 5 , with no less than 3 zeros) and find all their zeros (error less than 0.001), show your result and Matlab or Python code.

Problem 1 Solution

First, draw the graph of the higher degree equation of one variable then find the approximate interval between zero, Finally, the binary method is used to determine the value of zero.

Following are our target functions:

$$y = x^6 + x^4 - 3x^5 - 2x^3 + 4x^2 \quad (1)$$

$$y = x^7 - 2x^6 - 4x^5 + 3x^3 + 3x^2 \quad (2)$$

Problem 2(1)

Find the numerical solution of the following nonlinear equations using iteration method and Newton method respectively and show your analysis of error by using norms of vector and matrix and demonstrate your computer code

$$\begin{cases} x_1^2 - 10x_1 + x_2^2 + 8 = 0 \\ x_1x_2^2 + x_1 - 12x_2 + 7 = 0 \end{cases} \quad (3)$$

Problem 2(2)

Find the numerical solution of the following nonlinear equations using iteration method and Newton method respectively and show your analysis of error by using norms of vector and matrix and demonstrate your computer code

$$\begin{cases} 5x_1 - x_2^2 + x_1x_3 = 2 \\ x_1x_3 + 4x_2 - x_3 = 0 \\ x_1^2 + x_2 - 3x_3 = 1 \end{cases} \quad (4)$$

x p $f(x)$

$$[x, p] f(x) = \left[x \frac{\hbar}{i} \frac{d}{dx} (f) - \frac{\hbar}{i} \frac{d}{dx} (xf) \right] = \frac{\hbar}{i} \left(x \frac{df}{dx} - x \frac{df}{dx} - f \right) = i\hbar f(x) \quad (5)$$

$$[x, p] = i\hbar \quad (6)$$

(11)

$$a_- a_+ = \frac{1}{\hbar\omega} H + \frac{1}{2} \quad (7)$$

a_+

$$a_+ a_- = \frac{1}{\hbar\omega} H - \frac{1}{2} \quad (8)$$

$$[a_-, a_+] = 1 \quad (9)$$

(14)(15)

$$H = \hbar\omega(a_-a_+ - \frac{1}{2}) \quad (10)$$

$$H = \hbar\omega(a_+a_- + \frac{1}{2}) \quad (11)$$

$a_{\pm},$

$$\hbar\omega(a_{\pm}a_{\mp} \pm \frac{1}{2})\psi = E\psi \quad (12)$$

$$\psi \quad E \quad H\psi = E\psi, \quad a_+\psi \quad (E + \hbar\omega) \quad H(a_+\psi) = (E + \hbar\omega)(a_+\psi)$$

$$\begin{aligned}
 H(a_+\psi) &= \hbar\omega(a_+a_- + \frac{1}{2})(a_+\psi) \\
 &= \hbar\omega(a_+a_-a_+ + \frac{1}{2}a_+)\psi \\
 &= \hbar\omega a_+(a_-a_+ + \frac{1}{2})\psi \\
 &= a_+ \left[\hbar\omega(a_+a_- + 1 + \frac{1}{2})\psi \right] \\
 &= a_+(H + \hbar\omega)\psi = a_+(E + \hbar\omega)\psi = (E + \hbar\omega)(a_+\psi)
 \end{aligned} \tag{13}$$

$$a_-\psi \quad (E - \hbar\omega)$$

$$a_{\pm}$$

$$a_- \psi_0 = 0 \quad (14)$$

$$\psi_0(x)$$

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2} \quad (15)$$

$$(19) \quad \hbar\omega(a_+a_- + \frac{1}{2})\psi_0 = E_0\psi_0,$$

$$E_0 = \frac{1}{2}\hbar\omega \quad (16)$$

$$\psi_0,$$

$$\psi_n(x) = A_n(a_+)^n \psi_0(x)$$

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega \quad (17)$$

$$A_n$$

$$a_{\pm} \psi_{n\pm 1},$$

$$a_+ \psi_n = c_n \psi_{n+1} \quad , \quad a_- \psi_n = d_n \psi_{n-1} \quad (18)$$

$$f(x) g(x)$$

$$\int_{-\infty}^{+\infty} f^*(a_{\pm} g) dx = \int_{-\infty}^{\infty} (a_{mp} f)^* g dx \quad (19)$$

$$a_{\mp} a_{\pm}$$

$$\begin{aligned} \int_{-\infty}^{+\infty} f^*(a_{\pm} g) dx &= \frac{1}{\sqrt{2\hbar m\omega}} \int_{-\infty}^{+\infty} f^* \left(\mp \hbar \frac{d}{dx} + m\omega x \right) g dx \\ &= \frac{1}{\sqrt{2\hbar m\omega}} \int_{-\infty}^{+\infty} \left[\left(\pm \hbar \frac{d}{dx} + m\omega x \right) f \right]^* g dx \\ &= \int_{-\infty}^{+\infty} (a_{\mp} f)^* g dx \end{aligned} \quad (20)$$

$$\int_{-\infty}^{+\infty} (a_{\pm}\psi_n)^*(a_{\pm}\psi_n)dx = \int_{-\infty}^{+\infty} (a_{\mp}a_{\pm}\psi_n)^*\psi_n dx \quad (21)$$

(19) (24)

$$a_+a_-\psi_n = n\psi_n \quad , \quad a_-a_+\psi_n = (n+1)\psi_n \quad (22)$$

$$\begin{aligned} \int_{-\infty}^{+\infty} (a_+\psi_n)^*(a_+\psi_n)dx &= |c_n|^2 \int_{-\infty}^{+\infty} |\psi_{n+1}|^2 dx = (n+1) \int_{-\infty}^{+\infty} |\psi_n|^2 dx \\ \int_{-\infty}^{+\infty} (a_-\psi_n)^*(a_-\psi_n)dx &= |d_n|^2 \int_{-\infty}^{+\infty} |\psi_{n-1}|^2 dx = n \int_{-\infty}^{+\infty} |\psi_n|^2 dx \end{aligned} \quad (23)$$

$c_n \ d_n,$

$$a_+ \psi_n = \sqrt{n+1} \psi_{n+1} \quad a_- \psi_n = \sqrt{n} \psi_{n-1} \quad (24)$$

$$\psi_1 = a_+ \psi_0$$

$$\psi_2 = \frac{1}{\sqrt{2}} a_+ \psi_1 = \frac{1}{\sqrt{2}} (a_+)^2 \psi_0$$

$$\psi_3 = \frac{1}{\sqrt{3}} a_+ \psi_2 = \frac{1}{\sqrt{3 \times 2}} (a_+)^3 \psi_0$$

$$\psi_4 = \frac{1}{\sqrt{4}} a_+ \psi_3 = \frac{1}{\sqrt{4 \times 3 \times 2}} (a_+)^4 \psi_0$$

$$\psi_n = \frac{1}{\sqrt{n!}} (a_+)^n \psi_0 \quad (25)$$

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi$$

$$\xi \equiv \sqrt{\frac{m\omega}{\hbar}}x \quad (26)$$

$$\frac{d^2\psi}{d\xi^2} = (\xi^2 - K)\psi \quad (27)$$

$$K \equiv \frac{2E}{\hbar\omega} \quad (28)$$

ξ K

$$\frac{d^2\psi}{d\xi^2} \approx \xi^2\psi \quad (29)$$

$$\psi(\xi) \approx Ae^{-\frac{\xi^2}{2}} + Be^{\frac{\xi^2}{2}} \quad (30)$$

B

$$\psi(\xi) = h(\xi)e^{-\frac{\xi^2}{2}} \quad (31)$$

$$\begin{aligned}\frac{d\psi}{d\xi} &= \left(\frac{dh}{d\xi} - \xi h\right)e^{-\frac{\xi^2}{2}} \\ \frac{d^2\psi}{d\xi^2} &= \left(\frac{d^2h}{d\xi^2} - 2\xi\frac{dh}{d\xi} + (\xi^2 - 1)h\right)e^{-\frac{\xi^2}{2}} \\ \frac{d^2h}{d\xi^2} - 2\xi\frac{dh}{d\xi} + (K - 1)h &= 0\end{aligned}\tag{32}$$

ξ

$$h(\xi) = a_0 + a_1\xi + a_2\xi^2 + \cdots = \sum_{j=0}^{\infty} a_j \xi^j \quad (33)$$

$$\frac{dh}{d\xi} = a_1 + 2a_2\xi + 3a_3\xi^2 + \cdots = \sum_{j=0}^{\infty} j a_j \xi^{j-1}$$

$$\frac{d^2 h}{d\xi^2} = 2a_2 + 2 \times 3a_3\xi + 3 \times 4a_4\xi^2 + \cdots = \sum_{j=0}^{\infty} (j+1)(j+2)a_{j+2}\xi^j$$

(39)

$$\sum_{j=0}^{\infty} [(j+1)(j+2)a_{j+2} - 2ja_j + (K-1)a_j] \xi^j = 0 \quad (34)$$

ξ

$$(j+1)(j+2)a_{j+2} - 2ja_j + (K-1)a_j = 0$$

$$a_{j+2} = \frac{(2j+1-K)}{(j+1)(j+2)}a_j \quad (35)$$

 a_0

$$a_2 = \frac{(1-K)}{2}a_0, a_4 = \frac{5-K}{12}a_2 = \frac{(5-K)(1-K)}{24}a_0, \dots$$

 a_1

$$a_3 = \frac{(3-K)}{6}a_1, a_5 = \frac{7-K}{20}a_3 = \frac{(7-K)(3-K)}{120}a_1, \dots$$

$$h(\xi) = h_{\text{even}}(\xi) + h_{\text{odd}}(\xi) \quad (36)$$

$$h_{\text{even}}(\xi) \equiv a_0 + a_2\xi^2 + a_4\xi^4 + \dots$$

$$h_{\text{odd}}(\xi) \equiv a_1\xi + a_3\xi^3 + a_5\xi^5 + \dots$$

j

$$a_{j+2} \approx \frac{2}{j} a_j \quad (37)$$

$$a_j \approx \frac{C}{(j/2)!} \quad (38)$$

$$h(\xi) \approx C \sum \frac{1}{j!} \xi^j \approx C \sum \frac{1}{j!} \xi^{2j} \approx e^{\xi^2} \quad (39)$$

$$(38) \quad \psi = e^{\frac{\xi^2}{2}}, \quad j = n,$$

$$K = 2n + 1 \quad (40)$$

$$(35) \quad \begin{matrix} n \\ a_1 = 0, \end{matrix} \quad \begin{matrix} n \\ a_0 = 0 \end{matrix}$$

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega \quad n = 1, 2, \dots$$

K ,

$$a_{j+2} = \frac{-2(n-j)}{(j+1)(j+2)} a_j \quad (41)$$

n=0

$$h_0(\xi) = a_0$$

$$\psi_0(\xi) = a_0 e^{-\frac{\xi^2}{2}}$$

n=1

$$h_1(\xi) = a_1 \xi$$

$$\psi_1(\xi) = a_1 \xi e^{-\frac{\xi^2}{2}}$$

n=2

$$h_2(\xi) = a_0(1 - 2\xi^2)$$

$$\psi_2(\xi) = a_0(1 - 2\xi^2) e^{-\frac{\xi^2}{2}}$$

$$H_n(\xi),$$

$$H_0 = 1$$

$$H_1 = 2\xi$$

$$H_2 = 4\xi^2 - 2$$

$$H_3 = 8\xi^3 - 12\xi \tag{42}$$

$$H_4 = 16\xi^4 - 48\xi^2 + 12$$

$$H_5 = 32\xi^5 - 160\xi^3 + 120\xi$$

$$H_n(\xi) = (-1)^n e^{\xi^2} \left(\frac{d}{d\xi} \right)^n e^{-\xi^2} \quad (43)$$

$$H_{n+1}(\xi) = 2\xi H_n(\xi) - 2n H_{n-1}(\xi) \quad (44)$$

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\frac{\xi^2}{2}} \quad (45)$$